

Roman telescope passes steady-pointing tests and sees its first stars

From: ScienceDaily Top Technology Mined: 2026-10-03 3 min read

Taking a sharp picture of space requires a very steady telescope. NASA's Nancy Grace Roman Space Telescope has now passed important tests of the system that helps it stay pointed at its targets.

From September 15 to 21, engineers tested Roman's fine-guidance system. It watches **guide stars**, whose positions are already known accurately. About four times each second, it reports their positions to the system that controls the telescope's direction. This lets Roman correct tiny movements while collecting light.

The telescope stayed steady to better than one hundred-thousandth of a degree. That is roughly like keeping a laser pointed at a U.S. dime from about 240 kilometers away!

On September 22, Roman's **Coronagraph** Instrument received light from space for the first time. A **coronagraph** reduces a star's bright light so scientists can study much fainter planets and dusty disks around stars. Even tiny movements could let unwanted starlight spoil an image.

The first test looked at a faint star in the Large Magellanic Cloud, a neighboring **galaxy**. Its **detectors** were kept warmer than their final working temperature to help prevent **contamination** from sticking to them. This made the image noisier.

For a second observation on September 27, engineers cooled the **detectors** and saw multiple stars. These were early checks of focus and pointing. More complicated tests are still needed before the instrument's future science observations.

Word Treasure

guide stars -- Stars with accurately known positions that help a telescope keep its direction.

coronagraph -- An instrument that reduces bright starlight so fainter objects can be studied.

detectors -- Parts of an instrument that sense light.

contamination -- Unwanted material that can stick to an instrument.

galaxy -- A large collection of stars.

Background you need

A steady telescope can collect light while keeping its target in the same direction.
Bright starlight can make much fainter objects around a star difficult to study.

Break it down

Introduces Roman's successful steady-pointing tests.
Explains how guide stars help correct tiny movements.
Describes the coronagraph's purpose and first star observations.
Ends with the need for more complicated tests.

Why it matters

Steady pointing and focused images help prepare Roman to study faint objects in space.

Test what you remember

1. What does Roman's fine-guidance system watch?
 - (A) Galaxies with bright centers
 - (B) Clouds with changing shapes
 - (C) Stars with known positions
 - (D) Planets with dusty rings
2. How often does the system report the guide stars' positions?
 - (A) About four times each second
 - (B) About four times each minute
 - (C) About four times each hour
 - (D) About four times each day
3. What does a coronagraph reduce?
 - (A) A telescope's collecting time
 - (B) A star's bright light
 - (C) A galaxy's apparent size
 - (D) A detector's working temperature
4. Where was the faint star in the first test?
 - (A) In the telescope's guidance system
 - (B) In a dusty disk around a star
 - (C) In the Large Magellanic Cloud
 - (D) In a group of nearby planets
5. Why were the detectors kept warmer during the first test?
 - (A) To help increase image noise
 - (B) To help reduce bright starlight
 - (C) To help correct tiny movements
 - (D) To help prevent contamination
6. What did engineers see after cooling the detectors for the second observation?
 - (A) Multiple stars
 - (B) Multiple nearby galaxies
 - (C) Multiple dusty disks
 - (D) Multiple planets

Answer key (try the quiz first!): 1: C 2: A 3: B 4: C 5: D 6: A

MY THOUGHTS (at least 20 words):